MIL-L-18937D
13 May 1976
SUPERSEDING
MIL-L-18937C
2 February 1971

MILITARY SPECIFICATION

LANTERN, ELECTRIC, HAND, SAFETY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers some types of safety electric hand lanterns using standard commercial 6-volt batteries designed for use in class I, group D hazardous locations, as defined by NFPA No. 70.

1.2 Classification. Lanterns shall be of the following types, styles, and sizes as specified (see 6.2).

Type 1 - Metal lantern case.
Type 2 - Plastic lantern case.

Style 1 - Cylindrical battery compartment with detached main lamp housing mounted on the longitudinal side of battery compartment.
Style 2 - Rectangular battery compartment with detached main lamp housing mounted above battery compartment.

Size 1 - One battery capacity.
Size 2 - Two battery capacity.

Style 3 - Battery compartment integral with lamp housing.
Style 4 - Open battery compartment designed for use with clamp-on type batteries.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-101 - Battery, Dry.
W-L-200 - Lantern, Electric.

FSC 6230

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MILITARY

MIL-B-18 - Batteries, Dry.
MIL-B-18/34 - Battery, Dry, BA-200/U.
MIL-F-3222 - Floodlights and Lanterns, Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

C18.1 - Specifications for Dry Cells and Batteries.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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UNDERWRITERS' LABORATORIES, INC. (UL)

UL 783 - Standard for Electric Flashlights and Lanterns for use in Hazardous Locations Class I, Groups C and D.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The lantern shall be a complete unit including case, battery compartment, lamp housing, switch, carrying bail or handle, and such other parts necessary to provide a lantern suitable for use in hazardous areas. This lantern shall meet the minimum requirements of Standard UL 783 for class I, group D hazard locations as defined by NFPA No. 70, and other requirements specified herein. The label or listed markings of UL will be accepted as evidence that the lantern conforms to the requirements cited above.

3.2 First article. When specified (see 6.2), the contractor shall furnish one lantern for first article inspection and approval (see 4.3 and 6.4). The first article may be a standard production item from the contractor's current inventory provided the lantern meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining lanterns to be furnished under the contract.

3.3 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual component or the overall assembly.

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3.3.1 Metallic parts. All protruding metal parts shall be fabricated from either nonmagnetic metal or shall be adequately coated with a non-sparking material. All metal parts (other than current-carrying parts) shall be suitably protected against corrosion. When specified (see 6.2), all exterior metal parts shall be of nonmagnetic metals.

3.3.2 Plastic cases. Plastic cases shall be fabricated from acrylonitrile-butadiene-styrene (ABS) copolymer material similar or equal to Marborn Chemical Co., Cycolac, type T; or polycarbonate resin material similar or equal to General Electric Co., Lexan.

3.3.3 Enclosure. All exposed external surfaces of the lantern enclosure shall be made of nonsparking material, except that when the exposed surface or part is protected by a recess or by a guard, or when percussion sparks are unlikely, limited use of sparking material will be permitted in accordance with UL 783.

3.3.4 Dissimilar metals. The use of dissimilar metals, as defined in MIL-STD-889, in intimate contact with each other shall be avoided wherever possible in order to minimize galvanic corrosion between these metals. When the use of dissimilar metal cannot be avoided, an interposing material which will minimize or eliminate this galvanic effect shall be used.

3.4 Type 1 lanterns. Type 1 lanterns shall have lantern cases fabricated from metal with a thickness of not less than 0.025 inch. The case shall be of corrosion-resisting metal or a metal having a corrosion-resisting finish applied by plating or enameling both internally and externally.

3.5 Type 2 lanterns. Type 2 lanterns shall have lantern cases fabricated from ABS copolymer plastic material or polycarbonate resin plastic material with a thickness of not less than 0.047 inch.

3.6 Styles.

3.6.1 Style 1. Style 1 lanterns shall have a cylindrical battery-compartment with a detached main lamp housing. The main lamp housing shall be mounted on the longitudinal side of the battery compartment. An auxiliary lamp housing may be mounted on one end of the battery compartment.

3.6.1.1 Stand and carrying bail. Style 1 lanterns shall be provided with a sturdy stand capable of supporting a mass of 20 pounds (lb) without damage. The carrying bail shall be formed from either metal or plastic tubing with a nominal diameter of not less than 3/8 inch. The stand and carrying bail shall be pivoted at or near the center of gravity

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of the lantern, with battery installed, so as to permit the lantern to be rotated through a vertical arc of not less than 245° while resting on its stand. The stand and carrying bail shall be connected to the case at the point of pivot by a friction means using either adjusting nuts or spring tension to permit the lantern to be adjusted firmly in place.

3.6.2 Style 2. Style 2 lanterns shall have a rectangular battery compartment detached from the lamp housing. The lamp housing shall be mounted above the battery compartment. A size 1 lantern shall be designed to hold one lantern battery. A size 2 lantern shall be designed to hold two lantern batteries, and operated on either one or two batteries.

3.6.3 Style 3. Style 3 lanterns shall have the battery compartment integral (see 6.7.2) with the lamp housing.

3.6.4 Style 4. Style 4 lanterns shall have an open battery compartment designed to clamp on a single 6-volt battery of the type specified in 3.10.

3.7 Lamp housing. Lanterns shall be provided with one or two lamp housings sufficient to accommodate reflector(s) using either a single lamp or twin lamp socket with brass spring contacts. The lamp housing shall contain the lamp, lampholder, lamp ejecting mechanism, reflector, and lens.

3.7.1 Lamp housing movement. Style 1 lanterns shall have a main lamp housing capable of movement as specified in 3.6.1.1. The main lamp housing of styles 2, 3, and 4 lanterns shall be capable of being rotated through a vertical angle of not less than 90° measured from a normal plane, inclined 10° below the horizontal, when the lantern is resting on its base in a horizontal plane. The main lamp housing shall remain as positioned after being rotated through any angle within its range of adjustment. The main lamp housing and lantern case may move together, or relative to each other, in order to provide for the required movement.

3.7.2 Lamp. The lamp shall be a miniature, clear lamp designed to operate with a nominal 6-volt battery. Lamp current shall be less than 0.6 ampere. Unless otherwise specified (see 6.2), the type of lamp shall be the manufacturer's standard lamp.

3.7.3 Lamp socket. The lamp socket shall locate and retain a miniature clear lamp at the focal point of the reflector. The socket shall have the additional capability of lamp retention and expulsion as required to qualify for use in class I, group D hazardous location areas.

3.7.4 Spare lamps. Two spare lamps shall be supplied in a spare lamp clip mounted inside the lantern.

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3.7.5 Reflector. The reflector shall be suitably plated and polished to a high reflecting nontarnishing surface. The effective diameter of the reflector shall be not less than 2.5 inches.

3.7.6 Lens. The reflector and lamp socket shall be protected from dirt and moisture by a clear lens of either polycarbonate or alkyd resin transparent thermoplastic material, or of glass adequately tempered for shatter resistance. The lens shall be free from all defects or blemishes which would affect the light distribution.

3.8 Handle. A sturdy hand carrying handle or carrying bail shall be attached to the case. The carrying bail shall distribute the weight of the lantern without excessive strain on any part.

3.9 Base. The lanterns shall have a suitable base, stand, or surface intended by the manufacturer to serve as the support or bottom of the lantern. The base shall be sturdy and capable of supporting a mass of 20 lb without damage.

3.10 Batteries. Lanterns shall be designed to accommodate batteries that conform to ANSI C18.1, W-B-101, and MIL-B-18, as applicable. Batteries shall be the National Electronic Distributors Association (NEDA) types indicated in Table I. The battery(s) shall be held secure in the lantern.

TABLE I. Lantern battery types.

Lantern style	NEDA battery type	Military type conforming to MIL-B-18
1, 2, 3	908	MIL-B-18/34, type BA-200/U
4	918	

3.11 Light projection. The lantern shall be capable of projecting a concentrated beam of light, with maximum luminous intensity at approximately the center of the beam. The average beam candlepower shall be not less than 300 when measured with an illuminometer (see 4.6.1). The beam spread shall be such that the diameter of the concentrated spot of light shall be not greater than 20 inches when the lamp is at the focal point of the reflector, and the beam is projected on a screen mounted perpendicular to the axis of the beam at a distance of 10 feet from the lantern. The dispersed light surrounding the beam shall be distributed evenly through an included angle of approximately 200° about the axis of the beam.

3.12 Electrical connections. Except where otherwise specified herein, electrical connections shall be mechanically and electrically secured by either spring contact, soldering, riveting, or crimping. Connections for the battery spring contacts shall be adequate to cover normal variations in the location of battery springs.

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3.13 Switch. The lantern shall be provided with a switch that is suitable for not less than 25,000 operations. When the lantern is equipped with a twin lamp socket, a switching mechanism shall be provided to enable either lamp to be moved to the focal point of the reflector and lighted. Switch operation shall conform to UL 783.

3.14 Carrying strap. When specified (see 6.2), a carrying strap shall be furnished. Means for attaching the strap to the lantern shall be provided on the lantern. Installation of the carrying strap shall not necessitate the removal of the carrying handle.

3.15 Instructions. Instructions for lamp and battery installation shall be permanently affixed inside the battery compartment. Instructions shall be embossed, stenciled, or printed.

3.16 Treatment and finishing. The lantern shall be treated and finished in accordance with the manufacturer's best standard practice.

3.17 Identification marking. The complete lantern assembly shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed on one lantern when a first article sample is required (see 3.2). This inspection shall include the examination of 4.5 and tests of 4.6. The first article may be a standard production item from the contractor's current inventory provided the lantern meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining lanterns to be furnished under the contract.

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4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the test of 4.6.

4.4.1 Inspection lot. A lot shall consist of all lanterns of the same description offered for delivery to the Government at one time under a specific contract.

4.4.2 Sampling for examination. A random sample of lanterns shall be selected from each lot in accordance with inspection level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 4.0 percent defective.

4.4.3 Sampling for tests. A random sample of lanterns shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-3. The AQL shall be 2.5 percent defective.

4.5 Examination. Each sample selected in accordance with 4.4.2 shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.6 Tests.

4.6.1 Beam candlepower test. Each lantern in the sample selected in accordance with 4.4.3 shall be tested to determine conformance to the beam candlepower requirements of 3.11. The lantern shall be operated with the specified lamp and battery in place, at a distance of 6 feet from the test plate of a standard illuminometer (Macbeth or equal), equippe with a lens to bring into focus and restrict the test field. The axis of the light beam shall be perpendicular to the surface of the test plate. A total of six readings of the luminous intensity value of the light beam shall be recorded and the average of the luminous intensity value shall be determined. The beam candlepower shall be calculated by the following formula.

$$\text{Beam candlepower} = 36 \text{ feet}^2 \times \text{average luminous intensity (foot-candles.)}$$

4.6.2 Hazardous location test. Each lantern in the sample selected in accordance with 4.4.3 shall be tested to determine conformance to the requirement of 3.1. The lantern shall be successfully tested for conformance to the requirements of UL 783, as specified in 3.1.

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4.7 Packaging inspection. The preservation-packaging, packing, and marking of the lanterns shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The lanterns shall be preserved and packaged in accordance with level A of MIL-F-3222.

5.1.2 Level C. The lanterns shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. This level may conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The lanterns shall be packed in accordance with level A of MIL-F-3222.

5.2.2 Level B. The lanterns shall be packed in accordance with level B of MIL-F-3222.

5.2.3 Level C. The complete lanterns shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lantern is intended to provide illumination in confined areas where flammable gases or vapors exist in sufficient quantity to render the atmosphere explosive or ignitable. See W-L-200 for lanterns intended for general use in nonhazardous locations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 1.2).

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(c) When a first article is required for inspection and approval (see 3.2, 4.3, and 6.4).
(d) Whether nonmagnetic metal construction is required (see 3.3.1).
(e) Type of lamp required, if other than manufacturer's standard (see 3.7.2).
(f) When a carrying strap is required (see 3.14).
(g) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisons of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the ASPR. The first article should be a first production item consisting of one lantern or it may be a standard production item from the contractor's current inventory as specified in 4.3. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, test, and approval of the first article.

6.5 Cross-reference of classifications. Revison D of this specification adds type, style, and size classifications (see 1.2).

Previous classification MIL-L-18937C	New classification MIL-L-18937D
none	Type 1, style 1

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensive changes involved.

6.7 Definitions.

6.7.1 Detached lamp housing. Detached lamp housing denotes the lamp housing and battery compartment are two separate self-contained units fastened mechanically together.

6.7.2 Integral lamp housing. Integral lamp housing denotes the lamp housing and battery compartment together form one unit in the closed lantern housing.

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DOCUMENT IDENTIFIER AND TITLE
MIL-L-18937D LANTERN, ELECTRIC, HAND, SAFETY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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